Output

```
UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 1
Enter character to transmit: A
Transmitted: A

UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 1
Enter character to transmit: B
Transmitted: B

UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 1
Enter character to transmit: C
Transmitted: C
```

```
UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 3
UART Buffer: A B C

UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 2
Received: A

UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 2
Received: B

UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 2
```

```
UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 2
Received: C

UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 3
Buffer is empty!

UART COMMUNICATION
1. Transmit Data
2. Receive Data
3. Display Buffer
4. Exit
Enter your choice: 4
```